

Chapter Four

KNOWLEDGE REPRESENTATION& REASONING - II
KNOWLEDGE REPRESENTATION AND INFERENCING UNSING FIRST ORDER LOGIC

Outline

- Limitations of Proposition Logic
- Power of predicate logic
- FOL function and predicate symbol
- Syntax and semantics of FOL
- FOL sentence
- Quantifier in FOL
- Property of Quantifier
- Free and bounded variable
- Well formed formula
- Normal forms
- Conversion into CNF
- Substitution
- **Inference Rules Involving Quantifiers**
- Unification
- Skolem functions
- Generalized modus ponens
- Generalized resolution
- Conversion into CNF
- Backward and forward chaining



Propositional Logic Limitations

- ▶ Some of the limitations of propositional logic includes
 - ▶ **Context dependency**: unlike natural language, where meaning depends on context, meaning in propositional logic is **context-independent**
 - ▶ *Example of statements which are not a propositional statement*
 - ▶ he is a student
 - ▶ All of them submit the requirement
 - ▶ **very limited expressive power**: unlike natural language, propositional logic has very limited expressive power
 - ▶ Example cannot say “**pits cause breezes in adjacent squares**” except by writing one sentence for each square
 - ▶ **It only represent declarative sentences**: Propositional logic is declarative (sentence always have truth value)
 - ▶ **Deals with only finite sentences**: propositional logic deals satisfactorily with finite sentences composed using *not, and, or, if . . . Then, iff*

Limitations....

▶ e.g., if there are 3 students A , B and C
taking $p = \text{“}A \text{ has red hat”}$,

$q = \text{“}B \text{ has red hat”}$ and

$r = \text{“}C \text{ has red hat”}$,

the formula “*there exists a student with a red hat*” may be modeled as $p \vee q \vee r$.

▪ on infinite models this may require infinite formulas;

▶ example, “*each natural number is even or odd*” has to be translated as $(p_0 \vee q_0) \wedge (p_1 \vee q_1) \wedge (p_2 \vee q_2) \wedge \dots$

where $p_0, p_1, p_2, \dots$ are even and $q_0, q_1, q_2, \dots$ are odd. .

▶ The propositional logic assumes the world consists of **facts**.

▶ Cannot express the following:

All men are mortal

Socrates is a man

Therefore, Socrates is mortal

▶ Propositional Logic has thus limited expressive power.

Predicate Logic power

- ▶ Equivalent names
 - ▶ Predicate logic
 - ▶ First Order Logic (FOL)
 - ▶ First Order Predicate Calculus (FOPC).
- ▶ **Predicate logic** is an extension of propositional logic using variables for *objects*
- ▶ It is much **richer** and **complex** than propositional logic.
- ▶ **Predicate logic** has complex expressive power
- ▶ e.g., If **x** represents a natural number, then “**each natural number is even or odd**”; may be written shortly as
- ▶ **$\forall x(E(x) \vee O(x))$** where $E(x)$ = “ x is even” and $O(x)$ = “ x is odd”

FOL function and predicate symbols

- ▶ Function symbols are symbols that takes argument as a set of terms (variables, constant, functions) that represents a new object
 - ▶ Father(John)
 - ▶ Sucessor(X)
 - ▶ Sucessor(Sucessor(2))
- ▶ A ***predicate symbols*** is a symbol which describes a ***relation*** between objects or property of an object
 - ▶ Father(solomon, gizaw)
 - ▶ Male(teshome)

Example1

Let: $s(x)$ is the successor function and

$L(x,y)$ represents the relation “ $x < y$ ”,

The statement “a number is smaller than its successor ” can be represented as

$$\forall x L(x,s(x))$$

where domain is implicitly known

$$\forall x N(x) \rightarrow L(x,s(x))$$

where domain is explicitly mentioned

Example2

Represent the statement “Not all birds can fly”

Let $B(x)$ denotes “ x is a bird.”

Let $F(x)$: denotes “ x can fly”;

$$\sim(\forall x (B(x) \rightarrow F(x)))$$

Does this equivalent to

$$\exists x (B(x) \wedge \neg F(x))$$



Example3

Represent the statement “**All men are mortal. Socrates is man. Therefore, Socrates is Mortal**”

Let:

$H(x)$ denotes “ x is man”;

$M(x)$ denotes “ x is mortal ”, and

s denotes Socrates;

The statement (2) may be described as:

$$\forall x(H(x) \rightarrow M(x)), H(s) \vdash M(s)$$

Example4

Represent the statement “**Ann likes Mary’s brother**”

$$\exists x (B(x,m) \wedge L(a,x))$$

Every natural language sentence which is not semantically ambiguous can be represented using logic

Syntax & Semantics of FOPC

Sentence : Atomic Sentence | Sentence Connective Sentence | Quantifier Variable,... Sentence | $\sim$ Sentence | (Sentence)

AtomicSentence : Predicate(Term₁, Term₂,...,Term_n) | Term = Term

Term: Function(Term,...) | Constant | Variable

Connective : $\wedge$ | $\vee$ | $\Leftrightarrow$ | $\rightarrow$

Quantifier : $\forall$ | $\exists$

Constant : A | X₁ | KingJohn | ...

Variable : a | x | s | ...

Predicate : Before | HasColor | Raining | ...

Function : Mother | LeftLegOf | ...

Term

- A term is a logical expression that refers to an **object** .
- It is a name for a thing.
- There are three kinds of terms which allows us to name things in the world.:
 1. **Constant symbol**,
Example: John, Japan, Bacterium
 2. **Variable symbol**,
Example: x,y a, t,...
 3. **Function symbols**,
Example: f(f(x)); mother_of(John); LeftLegOf(*John*)
- Note: A term with no variables is called a **ground term**. For example John, father(solomon)

Equality (=)

- Two terms $term_1$ and $term_2$ are **equal** under a given interpretation if and only if $term_1$ and $term_2$ refer to the same object

Example1:

- If the object referred by $Father(John)$ and the object referred to by $Henry$ are the same, then
 $Father(John) = Henry$

Example2:

- Definition of *Sibling* in terms of *Parent*
$$\forall x, y \text{ Sibling}(x, y) \Leftrightarrow \{ \sim(x = y) \wedge \exists m, f \sim (m = f) \wedge \text{Parent}(m, x) \wedge \text{Parent}(f, x) \wedge \text{Parent}(m, y) \wedge \text{Parent}(f, y) \}$$

Predicate symbols

- **Predicate symbols:** are symbols that stand to show a relationship among terms or to indicate property of a term

For example:

On(A,B) to mean A is on B (relation between A & B)

Sister(Senait, Biruk) to mean Senait is sister of Biruk

Female(Azeb) to mean Azeb is female (property of azeb being female)

Here 'On', 'Sister' and Female are predicate symbols, '*A*', '*B*', '*Senait*', '*Biruk*' and *Azeb* are terms.

Atomic sentences

- Atomic sentences states the facts of world and is formed from a predicate symbol followed by a parenthesis list of terms.

Example:

Brother(Richard, John)

- Atomic sentences can have arguments that are complex terms:
 - ▶ *Sister(mother_of(John),Jane)*
 - ▶ *Married(FatherOf(Richard),MotherOf(John))*
- An atomic sentence is **true** if the relation referred to by the predicate symbol holds between the objects referred to by the arguments.

Complex Sentence

- Complex sentences are sentences which is a combination of one or more atomic sentences with logical connectives

Example:

- $\text{Older}(\text{John}, 30) \vee \text{Younger}(\text{John}, 30);$
represents John is above 30
- $\sim \text{Brother}(\text{Robin}, \text{John})$
represents Robin is Brother of John
- $\text{Hana} = \text{daughter}(\text{brother}(\text{mother}(\text{Selam}))) ;$
represents Hana is Selam's cousin
- $\text{Father}(\text{Solomon}, \text{Tesfaye}) \rightarrow \text{father}(\text{Solomon}, \text{Biruk})$
Represents if solomon is father of Tesfaye then he is also father of Biruk

Quantifiers

- A quantifier is a symbol that permits one to declare, or identify the **range** or **scope** of the variables in a logical expression.
- A quantifier express properties of entire collection of objects.
- There are two types of quantifier
 - Universal quantifier $\forall$
 - Existential quantifier $\exists$

Universal Quantification ($\forall$)

- Universal quantifier defines the domain of a variable in a logical expression to be the any element in the universe
- If x is a variable, then, $\forall x$ is read as
- **for all x** OR **for each x** OR **for every x**
- The scope of universal quantifier is the whole element in the domain

Syntax: $\forall <variables> <sentence>$

- one or more variables can be quantified by a single quantifier by separating with comma

Example1: “Every student is smart:”

$$\forall x (\text{Student}(x) \rightarrow \text{Smart}(x))$$

Example2: “All cats are mammals”

$$\forall x (\text{cats}(x) \rightarrow \text{Mammals}(x))$$

Universal Quantification ($\forall$)

- Roughly speaking, $\forall$ is equivalent to the *conjunction* of an *instantiations* of P
- $\forall x (\text{Student}(x) \rightarrow \text{Smart}(x))$ is equivalent to
 $(\text{Student}(\text{KingJohn}) \rightarrow \text{Smart}(\text{KingJohn})) \wedge$
 $(\text{Student}(\text{Abera}) \rightarrow \text{Smart}(\text{Abera})) \wedge$
 $(\text{Student}(\text{MyDog}) \rightarrow \text{Smart}(\text{MyDog})) \wedge \dots$
- $\forall x (\text{cats}(x) \rightarrow \text{Mammals}(x))$ is equivalent to
 $(\text{Cat}(\text{Spot}) \rightarrow \text{Mammals}(\text{Spot})) \wedge$
 $(\text{Cat}(\text{Rebecca}) \rightarrow \text{Mammals}(\text{Rebecca})) \wedge$
 $(\text{Cat}(\text{Felix}) \rightarrow \text{Mammals}(\text{Felix})) \wedge$
 $(\text{Cat}(\text{John}) \rightarrow \text{Mammals}(\text{John})) \wedge \dots$

Universal Quantification ($\forall$)

- Note1: Typically, $\rightarrow$ is the main connective with $\forall$
- Note2: **Avoid** the mistake of using $\wedge$ as the main connective with $\forall$:

Example:

- $\forall x (\text{Student}(x) \wedge \text{Smart}(x))$ means “Everyone is a student and everyone is smart”
- $\forall x (\text{cats}(x) \wedge \text{Mammals}(x))$ means “Everything is a cat and every thing is mammal”
- This doesn't agree in concept with the original sentence which is every student is smart, every cat are mammal respectively

Existential quantification ($\exists$)

- Existential quantifier defines the domain of a variable in a logical expression to be a non empty set which is the subset of the universal set
- if y is a variable, then $\exists y$ is read as
there exists a y OR for some y OR for at least one y

Syntax: $\exists$ <variables> <sentence>

- one or more variables can be quantified by a single quantifier by separating with comma

Example1: “Some students are smart”
 $\exists x (\text{student}(x) \wedge \text{Smart}(x))$

Example2: “Spot has a sister who is a cat”
 $\exists x (\text{Sister}(x, \text{Spot}) \wedge \text{Cat}(x))$

Existential quantification ($\exists$)

- Roughly speaking, $\exists$ is equivalent to the **disjunction** of **instantiations** of P
- $\exists x(\text{student}(x) \wedge \text{Smart}(x))$ is equivalent to
$$(\text{Student}(\text{KingJohn}) \wedge \text{Smart}(\text{KingJohn})) \vee$$
$$(\text{Student}(\text{Abera}) \wedge \text{Smart}(\text{Abera})) \vee$$
$$(\text{Student}(\text{MyDog}) \wedge \text{Smart}(\text{MyDog})) \vee \dots$$
- $\exists x(\text{Sister}(x, \text{Spot}) \wedge \text{Cat}(x))$ is equivalent to
$$(\text{Sister}(\text{Felix}, \text{Spot}) \wedge \text{Cat}(x)) \vee$$
$$(\text{Sister}(\text{Rabicca}, \text{Spot}) \wedge \text{Cat}(x)) \vee$$
$$(\text{Sister}(\text{chichu}, \text{Spot}) \wedge \text{Cat}(x)) \vee \dots$$

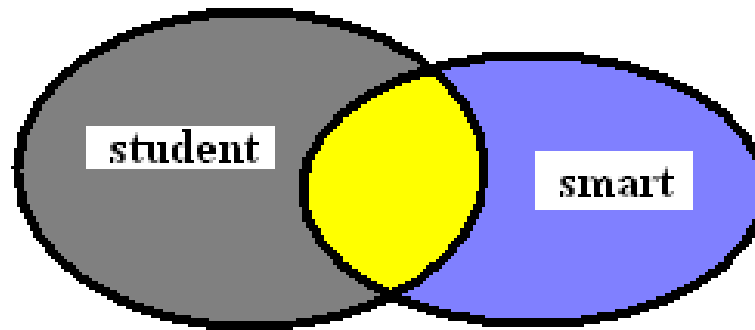
Existential quantification ($\exists$)

Note1: Typically, $\wedge$ is the main connective with $\exists$.

Note2: **Avoid** the mistake of using $\rightarrow$ as the main connective with $\exists$:

Example:

Some students are smart is to mean that
there are entities that satisfy the property of both being a student and smart.
In the figure, the yellow part indicates the set of such groups



However, $\exists x(\text{student}(x) \rightarrow \text{Smart}(x))$ is to mean any thing which is either *smart* or *not a student*

Hence this doesn't infer what we need to say

Examples using Quantifiers

- $\text{Man}(\text{John})$
John is a man
- $\forall x(\text{Man}(x) \rightarrow \sim \text{Woman}(x))$
Every man is not a woman (there is no man who is a woman)
- $\exists x(\text{Man}(x) \wedge \text{Handsome}(x))$
Some man are handsome
- $\forall x(\text{Man}(x) \rightarrow \exists y(\text{Woman}(y) \wedge \text{Loves}(x,y)))$
Every man has a woman that he loves
- $\exists y(\text{Woman}(y) \wedge (\forall x)(\text{Man}(x) \rightarrow \text{Loves}(x,y)))$
There are some woman that are loved by every man

Nested quantifiers

- Universal and existential quantifiers can be nested one into another.
- It is possible to have one or more quantifier nested in another quantifier

Example1

- “For all x and all y, if x is the parent of y then y is the child of x” can be represented as
$$\forall x, y (Parent(x, y) \rightarrow Child(y, x))$$
- Note: Here $\forall x, y$ means $\forall x \forall y$

Example2

- “There is someone who is loved by everyone”
$$\exists y \forall x Loves(x, y)$$

Nested quantifiers

- Difficulty may arise when two quantifiers are used with the same variable name.
- For example
$$\forall x (Cat(x) \vee \exists x Brother(Richard, x))$$
- in the sentence above, **x** in Brother(Richard,**x**) is existentially qualified and the universal quantifier has no effect on it.
- **Rule:** To identify which quantifier quantify a variable if the variable is quantified by two or more of them, the **innermost** quantifier that mentions it will be chosen.

Examples of Translating English to FOL

1. Every gardener likes the sun.

$$(\forall x) (\text{gardener}(x) \rightarrow \text{likes}(x, \text{sun}))$$

2. All purple mushrooms are poisonous

$$(\forall x) [(\text{mushrooms}(x) \wedge \text{purple}(x)) \rightarrow \text{poisonous}(x)]$$

3. No purple mushroom is poisonous

$$\sim(\exists x) (\text{purple}(x) \wedge \text{mushroom}(x) \wedge \text{poisonous}(x))$$

4. There are exactly two purple mushrooms

$$(\exists x)(\exists y) \{ \text{mushroom}(x) \wedge \text{purple}(x) \wedge \text{mushroom}(y) \wedge \text{purple}(y) \wedge \\ \sim(x=y) \wedge (\forall z)[(\text{mushroom}(z) \wedge \text{purple}(z)) \rightarrow ((x=z) \vee (y=z))]\}$$

Examples of Translating English to FOL

5. You can fool some of the people all of the time

$$(\exists x)(\text{person}(x) \wedge (\forall t) (\text{time}(t) \rightarrow \text{can-fool}(x,t)))$$

6. You can fool all of the people some of the time

$$(\forall x)(\text{person}(x) \rightarrow (\exists t) (\text{time}(t) \wedge \text{can-fool}(x,t)))$$

7. Jane is a tall surveyor

$$\text{Tall}(\text{Jane}) \wedge \text{Surveyor}(\text{Jane})$$

8. Everybody loves somebody

$$\forall x \exists y \text{ Loves}(x,y)$$

$$\exists y \forall x \text{ Loves}(x,y)$$

9. Nobody loves Jane

$$\forall x \sim \text{Loves}(x,\text{Jane})$$

$$\sim \exists y \text{ Loves}(y,\text{Jane})$$

Examples of Translating English to FOL

10. Everybody has a father

$$\forall x \exists y \text{ Father}(y,x)$$

11. Everybody has a father and mother

$$\forall x \exists y,z (\text{Father}(y,x) \wedge \text{Mother}(z,x))$$

12. Whoever has a father has a mother

$$\forall x \exists y (\text{Father}(y,x) \rightarrow \exists z \text{ Mother}(z,x))$$

13. Every son of my father is my brother

$$\forall x \forall y ((\text{MyFather}(x) \wedge \text{Son}(y,x)) \rightarrow \text{MYBrother}(y))$$

Properties of quantifiers (commutativity)

- i. $\forall x \forall y$ is the same as $\forall y \forall x$
- ii. $\exists x \exists y$ is the same as $\exists y \exists x$
- iii. $\exists x \forall y$ is **not** the same as $\forall y \exists x$,

Example

- ▶ $\exists x \forall y \text{ Loves}(x,y)$ means “There is a person who loves everyone in the world”
- ▶ $\forall y \exists x \text{ Loves}(x,y)$ means “Everyone in the world is loved by at least one person”

Properties of quantifiers (duality)

- **Quantifier duality** refers to the possibility of expressing one quantifier with the other equivalently
- Universal quantifier can be completely replaced by existential quantifier without affected the meaning and vice versa

Example

$$\forall x \text{ Likes}(x, \text{IceCream}) \equiv \neg \exists x \neg \text{Likes}(x, \text{IceCream})$$

$$\forall x \text{ Likes}(x, \text{IceCream}) \equiv \text{Everyone likes ice cream}$$

$$\neg \exists x \neg \text{Likes}(x, \text{IceCream}) \equiv \text{there is no one who does not like ice cream.}$$

$$\exists x \text{ Likes}(x, \text{Broccoli}) \equiv \neg \forall x \neg \text{Likes}(x, \text{Broccoli})$$

$$\exists x \text{ Likes}(x, \text{Broccoli}) \equiv \text{Some one likes Broccoli}$$

$$\neg \forall x \neg \text{Likes}(x, \text{Broccoli}) \equiv \text{It is not true that every one doesn't like Broccoli}$$

Properties of quantifiers (duality)

- Quantifiers are intimately connected with each other, through negation.

Example

- If one says that *everyone dislikes bitter guard* one is also saying that *there does not exists someone who likes them* or vice versa.
- $\forall$ is really conjunction over the universe of objects and $\exists$ is a disjunction over the universe, they obey De Morgan's rules.

Properties of quantifiers (duality)

The De Morgan rules for quantified and un-quantified sentences are as follows:

$$\forall x \sim P \equiv \sim \exists x P$$

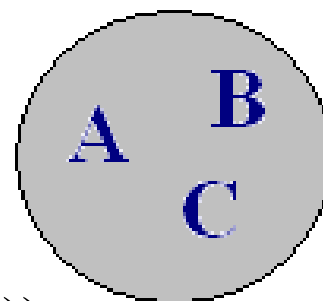
$$\sim \forall x P \equiv \exists x \sim P$$

$$\forall x P \equiv \sim \exists x \sim P$$

$$\exists x P \equiv \sim \forall x \sim P$$

In fact, one quantifier can do both works, if used with negation in appropriate place.

Consider a world consists of only three object A



Hence $\forall x P(x) \equiv (P(A) \wedge P(B) \wedge P(C))$

$$\equiv \neg \neg (P(A) \wedge P(B) \wedge P(C))$$

$$\equiv \neg (\neg P(A) \vee \neg P(B) \vee \neg P(C))$$

$$\equiv \neg (\exists x \neg P(x))$$

Syntax & Semantics of FOPC

$\forall x [\sim P(x)]$	P is false for all x
$\exists x [\sim P(x)]$	P is false for some x
$\forall x [P(x)]$	P is true for all x
$\exists x [P(x)]$	P is true for some x

Free and Bound variables

- A variable in a formula is **free** iff its *occurrence is outside the scope* of the qualifier having the variable.
- A variable in a formula is **bound** iff its *occurrence is within the scope* of the quantifier.
- The **scope** of a quantifier is the first *complete* sentence directly following the quantifier.

➤ Example1

$$\forall x P(x) \rightarrow \exists t Q(t) \wedge R(x,t)$$

- The scope of the variable x quantified by the universal quantifier is only in predicate P
- The scope of the variable t quantified by the existential quantifier is only in predicate Q
- The variables x and t in R are not quantified at all

$$\forall x (P(x) \rightarrow \exists t Q(t)) \wedge R(x,t)$$

Free and Bound variables

➤ Example2

$$\forall x (P(x) \rightarrow \exists t R(x,t)) \wedge Q(t)$$

- The scope of the variable x quantified by the universal quantifier is the preposition $P(x) \rightarrow \exists t R(x,t)$
- The scope of the variable t quantified by the existential quantifier is only in predicate R

Example-1:

- In the formula given below, the variables x and y are **free** and z is **bound** variables.

$$S(x,y) \wedge \forall z (P(z) \rightarrow Q(z)) \text{ ----(1)}$$

Free and Bound variables

Example-2:

- Consider the formula: $\forall x (A(x) \rightarrow B(x))$ ----(2)
- In this formula, the quantifier $\forall$ applies over the entire formula $(A(x) \rightarrow B(x))$.
- Hence the **scope** of the quantifier is entire $(A(x) \rightarrow B(x))$.
- A variable is **bound** in situation where at least one occurrence of it is bound.
- Similarly a variable is **free** in a formula, if all of its occurrence is free.

Example-3:

- In the formula $\forall x \exists y (A(x,y,z)) \wedge \forall z (B(y,z))$, ----(3)
- the variable z is free in the first portion of the formula and **bound** in the second portion of the formula.
- Hence the two occurrence of z are different variables

Well formed formula

➤ Well formed formula is a formula written in logical expression that can be interpreted perfectly into equivalent natural language sentence.

➤ we can use inductive definition to define wff as follows:

➤ Every atomic sentence is wff

➤ If W is wff then $\sim W$ is also wff

➤ If W and V are wff so are

- $W \wedge V$

- $W \rightarrow V$

- $W \vee V$

- $W \Leftrightarrow V$

➤ If W is wff so are

- $\exists x W$

- $\forall z W$

Normal Forms:

- A well formed formula can be represented in different standard normal forms
- Some of the normal forms are
 1. **Clause Form:** disjunction of literals (atomic sentences)
 2. **Conjunctive Normal Forms (CNF):** conjunction of disjunction of literals or atomic sentences. It can also be defined as conjunction of clauses
 3. **Disjunctive Normal Form (DNF):** disjunction of conjunction of literals.
 4. **Horn Form:** conjunction of literals (atomic sentences) implies a literal (atomic sentence).
 5. Others

Normal Forms:

Conjunctive Normal Form (CNF) is the focus here since:

1. any well formed formula (logical expression) can be converted into CNF
2. Generalized resolution is a complete inference procedure on CNF expression KB
3. It provides an easy way of inference procedure for the computer through resolution and refutation

A single literal (atomic sentence) or a single clause is in CNF form

$$Q \equiv Q \vee \text{False}$$

$$\sim Q \equiv \sim Q \vee \text{False}$$

► **Steps to convert from predicate logic formula to CNF.**

- 1) Eliminate implications and bi-conditionals.

$$(A \rightarrow B) = \sim A \vee B$$

$$(A \leftrightarrow B) = (A \rightarrow B) \wedge (B \rightarrow A)$$

- 2) Reduce the scope of negation and apply De Morgan's theorem to bring negations before the atoms

$$\sim(A \vee B) = \sim A \wedge \sim B$$

$$\sim(A \wedge B) = \sim A \vee \sim B$$

- 3) To bring the signs before the atoms, use the duality relation formulae

$$\sim \forall x (A(x)) = \exists x (\sim A(x))$$

$$\sim (\exists x (A(x))) = \forall x (\sim A(x))$$

- 4) For the sake of clarity (to avoid repetition) rename bound variables if necessary.

- 5) Use the equivalent formulae to move the quantifiers to the left of the formulae to obtain the normal form

Conjunctive Normal Forms

Formulae to be used in CNF conversion

- 1) $(Qx) A[x] \vee B \Leftrightarrow (Qx) (A[x] \vee B)$
- 2) $(Qx) A[x] \wedge B \Leftrightarrow (Qx) (A[x] \wedge B)$
- 3) $\sim(\forall x) (A[x]) \Leftrightarrow \exists x (\sim A[x])$
- 4) $\sim(\exists x) (A[x]) \Leftrightarrow \forall x (\sim A[x])$
- 5) $\forall x A[x] \wedge \forall x B[x] \Leftrightarrow (\forall x) (A[x] \wedge B[x])$
- 6) $\exists x A[x] \vee \exists x B[x] \Leftrightarrow (\exists x) (A[x] \vee B[x])$
- 7) $(Q_1 x) A[x] \vee (Q_2 x) B[x] \Leftrightarrow (Q_1 x) (Q_2 y) (A[x] \vee B[y])$
- 8) $(Q_1 x) A[x] \wedge (Q_2 x) B[x] \Leftrightarrow (Q_1 x) (Q_2 y) (A[x] \wedge B[y])$

Note 1. Q represents either $\forall$ or $\exists$

Note 2. $A[x]$ means the variable x is free in A

Note 3. B represents a formula that does not contain variable x.

Note 4. Q_1 and Q_2 are either $\forall$ or $\exists$. And y does not appear in $A[x]$.

Conversion exercise into CNF

1. $\forall x (A(x) \rightarrow \exists y B(x,y)) \text{ ---(1)}$

$$\equiv \forall x (\sim A(x) \vee \exists y B(x,y)) \quad \text{implication elimination}$$

$$\equiv \forall x \exists y (\sim A(x) \vee B(x,y)) \quad \text{pushing } \exists \text{ to the front}$$

- (solution)

2. $\exists x (A(x) \rightarrow \forall x B(x))$

$$\equiv \exists x (A(x) \rightarrow \forall y B(y)) \quad \text{variable renaming}$$

$$\equiv \exists x (\sim A(x) \vee \forall y B(y)) \quad \text{eliminating } \rightarrow$$

$$\equiv \exists x \forall y (\sim A(x) \vee B(y)) \quad \text{pushing } \forall \text{ to the front}$$

3. $\exists x A(x) \rightarrow \forall x B(x)$

$$\equiv (\exists x A(x)) \rightarrow \forall y B(y) \quad \text{scoping}$$

$$\equiv \sim(\exists x A(x)) \vee \forall y B(y) \quad \rightarrow \text{elimination}$$

$$\equiv \forall x \sim A(x) \vee \forall y B(y) \quad \text{pushing } \sim \text{ inward}$$

$$\equiv \forall x \forall y (\sim A(x) \vee B(y)) \quad \text{pushing } \forall \text{ to the front}$$

$$\equiv \forall x,y (\sim A(x) \vee B(y)) \quad \text{using single } \forall \text{ quantifier}$$

Conversion exercise into CNF

$$\begin{aligned} 4. \quad & \forall x \forall y [\forall z (A(x,y,z) \vee B(y)) \rightarrow \forall x C(x,z)] \\ & = \forall x \forall y [\forall z (A(x,y,z) \vee B(y)) \rightarrow \forall t C(t,s)] \\ & \quad \text{renaming the variable } x \text{ to } t, z \text{ to } s \text{ to avoid confusion} \\ & = \forall x \forall y [\sim \forall z (A(x,y,z) \vee B(y)) \vee \forall t C(t,s)] \\ & \quad \rightarrow \text{elimination} \\ & = \forall x \forall y [\exists z \sim (A(x,y,z) \vee B(y)) \vee \forall t C(t,s)] \\ & \quad \text{pushing } \sim \text{ inward} \\ & = \forall x \forall y [\exists z (\sim A(x,y,z) \wedge \sim B(y)) \vee \forall t C(t,s)] \\ & \quad \text{pushing } \exists \text{ to the front} \\ & = \forall x \forall y \exists z (\sim A(x,y,z) \wedge \sim B(y)) \vee \forall t C(t,s) \\ & \quad \text{renaming the free variable } z \text{ to } s \\ & = \forall x \forall y \exists z \forall t (\sim A(x,y,z) \wedge \sim B(y)) \vee C(t,s) \\ & \quad \text{pushing } \forall \text{ to the front} \\ & = \forall x \forall y \exists z \forall t (\sim A(x,y,z) \vee C(t,z)) \wedge (\sim B(y) \vee C(t,s)) \\ & \quad \text{distributive property of } \vee \text{ over } \wedge \end{aligned}$$

Substitution

Idea:

- if we replace a valid sentence with any constant uniformly, the new sentence is also valid. i.e.
- given a sentence with free variables, we can have multiple instance of the sentence by uniformly substituting one or more of its variables.
- This process is very important during inferencing by the agent which we will see soon.

Substitution

- Given a sentence S , free variable x , and a ground term t then we can represent a substitution of every occurrence of x by t in S as:
 $\text{SUBST}(\{x/t\}, S)$.
- e.g., $\text{SUBST}(\{x/f(x,y)\}, S(x,y) \wedge \forall x (P(x) \rightarrow Q(x))) \equiv \{ S(f(x,y),y) \wedge \forall x (P(x) \rightarrow Q(x)) \}$
- A sentence with free variable is called open sentence and a sentence with only bounded variable is called closed sentence.
- Substitution can be applied only on open sentence.
- Hence, it is important to know the principles how to make a variable in an expression free

Inference Rules Involving Quantifiers

- Those rules we have seen for propositional logic are also valid for FOPL
- A variable which is bound under the universal quantifier can be free if the world is clearly stated.
- For example: all normal human beings has two legs
- $\forall x(\textit{Normal-human}(x) \rightarrow \text{has-legs}(x, 2))$. Here x is bounded variable
- This can be written in known world as $\textit{Normal-human}(x) \rightarrow \text{has-legs}(x, 2)$. Here x is free variable

Inference Rules Involving Quantifiers

Example

Axioms: $\forall x (\text{Bird}(x) \rightarrow \text{Flies}(x))$

$\text{Bird}(\text{Tweety})$

Goal: $\text{Flies}(\text{Tweety})$

- By removing the universal quantifier and converting to clausal form, we will get

$\sim \text{Bird}(x) \vee \text{Flies}(x)$

$\text{Bird}(\text{Tweety})$

- Without further operation, no resolution is possible
- Apply **SUBST**($\{x/\text{Tweety}\}$, $\sim \text{Bird}(x) \vee \text{Flies}(x)$) gives

$\sim \text{Bird}(\text{Tweety}) \vee \text{Flies}(\text{Tweety})$

$\text{Bird}(\text{Tweety})$

- Using resolution, it results $\text{Flies}(\text{Tweety})$

Inference Rules Involving Quantifiers

- To help the agent for better inferencing, we have three more inference rules mentioned bellow:

1. Universal Elimination: for any sentence α variable x and ground term g :

$$\frac{\forall x, \alpha}{\text{SUBST}(\{x/g\}, \alpha)}$$

$\text{SUBST}(\{x/g\}, \alpha)$

example:

$\forall x \text{ Likes}(x, \text{IceCream})$

and infer

$\text{Likes}(\text{Helen}, \text{IceCream})$

we can use substitution $\{x/\text{Helen}\}$

Inference Rules Involving Quantifiers

- To help the agent for better inferencing, we have three more inference rules mentioned bellow:

2. Existential Elimination: for any sentence α variable x and constant symbol k that doesn't appear elsewhere in the knowledge base:

$$\frac{\exists x, \alpha}{\text{SUBST}(\{x/k\}, \alpha)}$$

example:

from $\exists x \text{ Kill}(x, \text{victim})$ we can infer

$\text{Kill}(\text{Murderer}, \text{victim})$ as long as Murderer doesn't appear in the KB.

Inference Rules Involving Quantifiers

3. Existential Introduction: for any sentence α variable x that doesn't occur in α and ground term g that occur in α :

$$\frac{\alpha}{\exists x, \text{SUBST}(\{g/x\}, \alpha)}$$

example:

from Likes(Jerry, IceCream)
 $\exists x \text{ Like}(x, \text{IceCream})$.

we can infer

Unification

- **Unification:** is a process of finding a substitution that makes two expressions match exactly each other.
- A substitution that makes two clauses resolvable is called a **unifier**.
- In other words, **unification** is a “pattern matching” procedure that takes two atomic sentences (*literals*), as input and returns:
 - “failure” if they do not match
 - a substitute list θ , if they do match.
- If p and q are two atomic sentences, a unification routine (UNIFY) takes p and q as arguments and returns a substitution that would make p and q look the same
- if there is no valid substitution it fails.
- $\text{UNIFY}(p, q) = \theta$ means
 $\text{SUBST}(\theta, p) = \text{SUBST}(\theta, q)$, where θ is called the **unifier** of the two sentences.

Unification

Example

- Assume the KB contains the following:

Knows(John,x) $\rightarrow$ Hates(John,x)

Knows(John, Jane)

Knows(y,Leonid)

Knows(y,Mothers(y))

Knows(x, Elizabeth)

Knows(solomon, mother(Jane))

- Ask the question $?Hates(John, z)$.
- Now, consult KB to know the ‘sentences’ that unify with $Knows(John, z)$ and then apply the unifier to Hates(John,z).

Unification

$\text{UNIFY}(\text{Knows}(\text{John}, z), \text{Knows}(\text{John}, \text{Jane})) = \{z/\text{Jane}\}$

$\text{UNIFY}(\text{Knows}(\text{John}, z), \text{Knows}(y, \text{Leonid})) = \{z/\text{Leonid}, y/\text{John}\}$

$\text{UNIFY}(\text{Knows}(\text{John}, z), \text{Knows}(y, \text{Mothers}(y))) = \{y/\text{John}, z/\text{mother}(\text{John})\}$

$\text{UNIFY}(\text{Knows}(\text{John}, z), \text{Knows}(x, \text{Elizabeth})) = \{x/\text{John}, z/\text{Elizabeth}\}$

$\text{UNIFY}(\text{Knows}(\text{John}, z), \text{Knows}(\text{solomon}, \text{mother}(\text{Jane}))) = \text{fail}$

Therefore, x can be Jane, Leonid, and Mother(John).

- Hence $\text{Hates}(\text{John}, x)$ can assume the values Jane, Leonid, and Mother(John).
- Thus a unifier is a function that assigns bindings to variables.
- A binding is either a constant, a function, or another variable

Generalized Modus-ponens

- For atomic sentences, p_i , p_i' and q where there is a substitution θ such that $\text{SUBST}(\theta, p_i') = \text{SUBST}(\theta, p_i)$, for all i :

$$\frac{p_1', p_2', \dots, p_n', (p_1 \wedge p_2 \wedge \dots \wedge p_n \rightarrow q)}{\text{SUBST}(\theta, q)}$$

- The generalized modes ponens is an efficient inference rule for three reasons:
1. Combines several small inferences into one
 2. It takes sensible steps (transitivity as a property and unification as a mechanism for matching substitution)
 3. We can represent knowledge in a **canonical form** during pre-compilation step (i.e. horn Normal form)

Generalized Modus-ponens

- Generalized Modus Ponens is **complete** for KBs containing only **Horn clauses**.
- A **Horn clause** is a sentence of the form:
$$(\forall_{vList}) P_1(vList) \wedge P_2(vList)(x) \wedge \dots \wedge P_N(vList) \rightarrow Q(vList)$$
 where zero or more P_i 's and Q which are positive literals (atomic sentences).
- Only Horn clause sentences are acceptable in prolog
- Horn clause represents a subset of the set of sentences representable in FOL.
- For example $R(x) \vee P(y) \vee Q(z)$ is a sentence in FOL, but not in Horn clause.

Resolution for FOPC

- The simplest resolution has exactly two disjunction in the premises (unit resolution)
- The generalized resolution extended the number of premises to any number of disjunctions

Generalized Resolution (disjunction):

- For literals (atomic sentences) p_i and q_i where $\text{UNIFY}(p_j, \sim q_k) = \theta$ then

$$p_1 \vee p_2 \vee \dots \vee p_j \vee \dots \vee p_m$$

$$q_1 \vee q_2 \vee \dots \vee q_k \vee \dots \vee q_n$$

$$\text{SUBST}(\theta, (p_1 \vee p_2 \vee \dots \vee p_{j-1} \vee p_{j+1} \vee \dots \vee p_m \vee q_1 \vee q_2 \vee \dots \vee q_{k-1} \vee q_{k+1} \vee \dots \vee q_n))$$

Resolution for FOPC

Proving theorem:

- **Method1:** Start with the given axiom, use the rules of inference and prove the theorem
- **Method2:** Prove that the negation of the result can not be TRUE.

Method2 is commonly known as prove by **refutation**.

Resolution for FOPC

*The **Resolution Refutation Procedure** involves **five** steps:*

1. Find the negation of the result to be proved
2. Add as a valid statement to the given set of statements (KB)
3. Perform the resolution on these statements until a contradiction is encountered.
4. Conclude that the contradiction is due to the assumed negation of the result
5. Hence, the negated assumption that is made is FALSE or the result is proved to be valid (TRUE).

First Order Resolution Examples

Example1:

Given

$$\forall x [\text{Physician}(x) \rightarrow \text{Knows_surgery}(x)]$$

$$\text{Physician}(\text{Biruk})$$

Prove $\text{knows_surgery}(\text{Biruk})$ using refutation method:

Assume the negation of the result is valid. That is

$$\sim \text{knows_surgery}(\text{Biruk}) \quad \text{--- (1)}$$

The given axioms are

$$\text{Physican}(\text{Biruk}) \quad \text{---- (2)}$$

$$\forall x [\text{Physician}(x) \rightarrow \text{Knows_surgery}(x)] \quad \text{----- (3)}$$

Drop the universal quantifier from equation (3)

$$\sim \text{Physician}(x) \vee \text{Knows_surgery}(x) \quad \text{---- (4)}$$

First Order Resolution Examples

In equation (4), substitute $x = \text{Biruk}$.

$$\sim \text{Physician}(\text{Biruk}) \vee \text{Knows_surgery}(\text{Biruk}) \text{ ---- (5)}$$

Resolving equations (1) and (5), we have

$$\begin{array}{l} \sim \text{Physican}(\text{Biruk}) \qquad \text{-----} \\ (6) \end{array}$$

- Resolving equations (2) and (6), we have contradiction.
- This contradiction was due to the assumption that was made at the beginning i.e., $\sim \text{knows_surgery}(\text{Biruk})$ is valid.
- Hence the assumption $(\sim \text{knows_surgery}(\text{Biruk}))$ is false which implies $\text{knows_surgery}(\text{Biruk})$ is true

First Order Resolution

- In FOL, to perform resolution, two main tasks are to be completed first:
1. writing expressions in Clausal form, and then
 2. Instantiate the variables in the universal statements by substitution and unification.

Example:

$\forall x (P(x) \rightarrow Q(x))$	;	All men are mortal
$P(A)$	;	Socrates is a man
_____ $Q(A)$	;	Socrates is mortal

Use equivalence rules to remove implication.

$\forall x (\sim P(x) \vee Q(x))$
$P(A)$
_____ $Q(A)$

Substitute A for x, the proposition remains still true.

$\sim P(A) \vee Q(A)$	
$P(A)$	
_____ $Q(A)$	;

Propositional resolution

Skolen Function

- A **Skolen function** eliminates existential qualifier($\exists$) and introduces a brand new constant symbol known as **Skolen constant** that is not used in any other sentence
- So to Skolemize means, to substitute a brand new name for each existentially quantified variable.

Example 1: (some one is a killer)

$\exists x \text{ Killer}(x)$ can be skolemized as ***Killer(Murderer)*** provided that the constant is a brand new constant

Example 2: (some one teaches some thing)

$\exists x, y \text{ Teaches}(x, y)$ can be skolemized as ***Teaches(Teacher, Course)*** provided that the constant is a brand new constant

Example 3: (there is some one who teaches every course)

$\exists x \forall y (\text{course}(y) \rightarrow \text{Teaches}(x, y))$ can be skolemized as ***$\forall y (\text{course}(y) \rightarrow \text{Teaches}(\text{Teacher}, y))$*** provided that the constant is a brand new constant

Skolen Function

Example 3: (every one teach some courses)

$\forall x \exists y (course(y) \wedge Teaches(x,y))$ is the following skolemization work:

$\forall x (course(TheCourse) \wedge Teaches(x, TheCourse))$ provided that the constant is a brand new constant

- No it doesn't work. Why?

Skolen Function

- If $\exists$ is in the scope of $\forall$ in the formula, then, $\exists$ is removed by replacing the variables as functions of variables preceding it.
- e.g., $\forall x \exists y \text{ Loves}(x,y)$ can be represented as $\forall x \text{ Loves}(x, \text{Beloved}(x))$
- But $\forall x \text{ Loves}(x, L)$ says every one loves the same person. But the original sentence says every one has some one to love.
- $\forall x (\text{Person}(x) \rightarrow \exists y \text{ Heart}(y) \wedge \text{Has}(x,y))$ Every one has a heart
- $\forall x (\text{Person}(x) \rightarrow \text{Heart}(H) \wedge \text{Has}(x,H))$. Every one has the same heart (shared)
- $\forall x (\text{Person}(x) \rightarrow \text{Heart}(f(x)) \wedge \text{Has}(x,f(x)))$. Where $f(x)$ is heart of x

Skolen Function

- **Example1:**
- Convert the formula into its Skolen Form

$$P = \exists q \forall r \exists s \exists t (A(q,r) \rightarrow B(s,t)) \text{ ---(1)}$$

Solution:

- In this formula, $\exists q$ is not preceded by any $\forall$ quantifier. Whereas $\exists s$ and $\exists t$ are in the scope of $\forall$. Hence replace q by a constant ' α ', and, s and t by functions of r .
- The Skolen form P^* of P is
$$P^* = \forall r (A(\alpha,r) \rightarrow B(f_1(r), f_2(r)))$$

Conversion of **FOL** to CNF by removing quantifier

- The following are guiding line for the conversion
 1. Eliminating implications and bi-implication
 2. Drive negative symbols inward
 3. Apply De Morgan laws on quantifiers
 4. Rename variables that may create ambiguity
 5. Skolemize existential quantifier
 6. Drop universal quantifiers
 7. Distribute $\wedge$ over $\vee$ which return clauses
 8. Rename the variables in each clause so that each clause will be distinct

FOL to CNF conversion Example

“Everyone who loves all animals is loved by someone”

Representation

Let $P(x)$ is *x loves all animal*

$Loves(y,x)$ is *y loves x*

Hence the original sentence is

$$\forall x (P(x) \rightarrow \exists y Loves(y,x))$$

$P(x) \equiv x$ loves all animal can be represented as

$$\forall y (Animal(y) \rightarrow loves(x,y))$$

Hence *every one who loves animal* can be represented as

$$\forall x P(x) \equiv \forall x (\forall y (Animal(y) \rightarrow loves(x,y)))$$

Therefore the entire expression becomes:

$$\forall x ((\forall y (Animal(y) \rightarrow Loves(x,y))) \rightarrow \exists y Loves(y,x))$$



FOL to CNF conversion Example

“Everyone who loves all animals is loved by someone”

$$\forall x ((\forall y (\text{Animal}(y) \rightarrow \text{Loves}(x,y))) \rightarrow \exists y \text{Loves}(y,x))$$

1. Eliminate bi-conditionals and implications

$$\forall x ((\forall y (\neg \text{Animal}(y) \vee \text{Loves}(x,y))) \rightarrow \exists y \text{Loves}(y,x))$$

$$\forall x (\neg (\forall y (\neg \text{Animal}(y) \vee \text{Loves}(x,y))) \vee \exists y \text{Loves}(y,x))$$

2. Move $\sim$ inwards: $\sim \forall x p \equiv \exists x \sim p$, $\sim \exists x p \equiv \forall x \sim p$

$$\forall x ((\exists y \neg (\neg \text{Animal}(y) \vee \text{Loves}(x,y))) \vee \exists y \text{Loves}(y,x))$$

$$\forall x ((\exists y (\text{Animal}(y) \wedge \neg \text{Loves}(x,y))) \vee \exists y \text{Loves}(y,x))$$

FOL to CNF conversion Example

3. Standardize variables: each quantifier should use a different one

$$\forall x ((\exists y (Animal(y) \wedge \sim Loves(x,y))) \vee \exists z Loves(z,x))$$

4. Skolemization: a more general form of existential instantiation. Each existential variable is replaced by a **Skolem function** of the enclosing universally quantified variables:

$$\forall x (((Animal(F(x)) \wedge \sim Loves(x, F(x)))) \vee Loves(G(x),x))$$

5. Drop universal quantifiers:

$$((Animal(F(x)) \wedge \sim Loves(x, F(x)))) \vee Loves(G(x),x)$$

6. Distribute $\vee$ over $\wedge$:

$$(Animal(F(x)) \vee Loves(G(x),x)) \wedge (\sim Loves(x, F(x)) \vee Loves(G(x),x))$$

First Order Resolution Example:

- Given the following premises, use resolution refutation procedure to prove that “curiosity killed Tuna”
 - a) John owns a dog
 - b) Any one who owns a dog is a lover-of-animals
 - c) Lovers-of-animals do not kill animals
 - d) Either Jack is killed Tuna or curiosity killed Tuna
 - e) Tuna is a cat
 - f) All cats are animals
- ***Knowledge representation in CNF***
 - a) John owns a dog
$$\exists x (D(x) \wedge O(J,x)) \quad \text{---(1)}$$
 - ***Skolemization*** . Since only existential qualifier is there in equ.(1), replacevariable x with a name, say “Fido”
$$D(\text{Fido}) \wedge O(J,\text{Fido}) \quad \text{---(2)}$$

First Order Resolution Example:

b) Anyone who owns a dog, is a lover of animals

$$\forall x (\exists y (D(y) \wedge O(x,y)) \rightarrow L(x)) \quad \text{--- (3)}$$

➤ Convert equ.(3) to Clausal form

$$\forall x (\sim \exists y (D(y) \wedge O(x,y)) \vee L(x))$$

➤ Drive negatives inside

$$\forall x (\forall y \sim (D(y) \wedge O(x,y)) \vee L(x))$$

$$\forall x (\forall y \sim D(y) \vee \sim O(x,y) \vee L(x))$$

➤ There is no need to perform Skolemization, as there are no existential qualifiers.

➤ Now, universal quantifiers can be dropped to get the required CNF:

$$\sim D(y) \vee \sim O(x,y) \vee L(x)$$

First Order Resolution Example:

c) Lovers-of-animals do not kill animals

$$\forall x (L(x) \rightarrow (\forall y (A(y) \rightarrow \sim K(x,y))))$$

$$\forall x (\sim L(x) \vee (\forall y (A(y) \rightarrow \sim K(x,y))))$$

$$\forall x (\sim L(x) \vee (\forall y (\sim A(y) \vee \sim K(x,y))))$$

$$\forall x \forall y (\sim L(x) \vee \sim A(y) \vee \sim K(x,y))$$

Drop universal quantifiers

$$\sim L(x) \vee \sim A(y) \vee \sim K(x,y)$$

d) Either Jack killed Tuna or curiosity killed Tuna

$$K(J,T) \vee K(C,T)$$

e) “Tuna is a cat

$$C(T)$$

f) All cats are animals

$$\sim C(x) \vee A(x)$$

First Order Resolution Example:

- The next step is to prove the theorem “curiosity killed Tuna” by refutation
- That is, we should assume **curiosity didn't killed Tuna** ($\sim K(C,T)$) as a valid sentence then the set of facts and rules that we have are:
 - a. $D(\text{Fido}) \wedge O(J, \text{Fido})$
 - b. $\sim D(y) \vee \sim O(x,y) \vee L(x)$
 - c. $\sim L(x) \vee \sim A(y) \vee \sim K(x,y)$
 - d. $K(J,T) \vee K(C,T)$
 - e. $C(T)$
 - f. $\sim C(x) \vee A(x)$
 - g. $\sim K(C,T)$

First Order Resolution Example:

Step No.	Clause	Derivation
1	$D(\text{Fido})$	a
2	$O(J, \text{Fido})$	a
3	$\sim D(y) \vee \sim O(x, y) \vee L(x)$	b
4	$\sim L(x) \vee \sim A(y) \vee \sim K(x, y)$	c
5	$K(J, T) \vee K(C, T)$	d
6	$C(T)$	e
7	$\sim C(x) \vee A(x)$	f
8	$\sim K(C, T)$	g Refutation
9	$K(J, T)$	Resolution of 5,8
10	$A(T)$	Resolution of 6,7 with unifier $\{x/T\}$
11	$\sim L(J) \vee \sim A(T)$	Resolution of 4,9 with unifier $\{x/J, y/T\}$
12	$\sim L(J)$	Resolution of 10,11
13	$\sim D(y) \vee \sim O(J, y)$	Resolution of 3, 12 with unifier $\{x/J\}$
14	$\sim D(\text{Fido})$	Resolution of 13, 2 with unifier $\{y/\text{Fido}\}$
15	*	Resolution of 14,1

► **Step-15 is a contradiction. Which proves that indeed curiosity killed the cat “Tuna”**